

Dilip Rana

PhD Candidate and Researcher at Rensselaer Polytechnic Institute

 ranad@rpi.edu [open](#)

 [linkedin.com/in/dilip-rana](https://www.linkedin.com/in/dilip-rana) [open](#)

 scholar.google.com/citations?user=hOYzsCcAAAAJ [open](#)

 [diliprana.github.io](https://github.io/diliprana) [open](#)

 github.com/DilipRana [open](#)

About me

I am a PhD candidate and work as a graduate research assistant in the field of energy and power electronics. My current research focuses on the reliability characterization of GaN power devices, with a particular emphasis on their operation at deep cryogenic temperatures (around 50 K). Under the supervision of Prof. Zheyu Zhang, I have learned to approach research with care, asking meaningful questions and tackling technical challenges with persistence. I look forward to further deepening my knowledge in the field of power electronics and power semiconductors.

Education

PhD in Electrical and Electronics Engineering (2024 – Expected graduation in 2029)

Rensselaer Polytechnic Institute, School of Engineering, Troy, New York

Term Average (until now): 3.88

Bachelor's Degree in Electrical Engineering (2017 - 2022)

Tribhuvan University, Institute of Engineering, Pulchowk Campus

Term Average: 74.99%

Higher Secondary Education, +2 Science (2015 - 2017)

Premier Higher Secondary School

Term Average: 73.29%

Experience

Graduate Research Assistant: Rensselaer Polytechnic Institute (Jan 2024 – Present)

- Involved in NASA funded project for performance evaluation of commercial GaN HEMTs
- Static characterization – extracting IV, transfer and leakage characteristics of commercial GaN HEMTs from major vendors at temperatures ranging from 50-400K
- Reliability characterization - gate reliability, application replication reliability and power cycling tests of selected GaN candidates
- Modeling of GaN HEMTs including statistical, dynamic and aging degradation behavior
- Supported all stages of lab operation and research by collaborating with postdoctoral and fellow lab members

Embedded Hardware Engineer: Yatri Motorcycles (Dec 2022 – Dec 2023)

- Prototyped and designed of different parts of the vehicle embedded system
- Performed signal integrity analysis of different boards designed and optimized the circuits
- Power management system designed with different filters with improved noise immunity and greater system reliability
- Designed, compared and tested of different methods of user input sensing in industry standard testing conditions
- Designed and implemented the various parts of the vehicle charging embedded system
- Supported all stages of embedded system development and collaborated with electrical and mechanical teams of the company

Embedded Hardware Engineer Intern: Yatri Motorcycles (Sep 2022 - Nov 2022)

- Completed the internship task within specified time and appreciated result
- Analyzed and designed noise shield for wiring of the vehicle

- Measured and verified different parasitic parameters of vehicle auxiliaries
- Designed and tested prototype of the noise filter for different noisy auxiliaries

Projects

Static Characterization of Commercial GaN Power Semiconductors (Jan 2024 - Jan 2025)

- Performed characterization of 14 commercial GaN HEMTs from major vendors including radiation hardened GaN parts for future space application
- Used cryocooler based setup to achieve temperature from deep cryogenic (50K) to elevated temperature (400K)
- Characterization of multiple devices considering statistical significance of obtained data, including concurrent characterization of 4 samples at once
- Extraction of temperature dependency of device performance parameters including – IV, transfer and leakage characteristics

Inductive Power Transfer: Electric Scooter based Design (Jun 2021 - Apr 2022)

- Performed 2D & 3D electromagnetic simulations and calculated the parameters of various coil designs using COMSOL Multiphysics
- Manually designed coil with stranded wires and matched up simulated parameters
- Half bridge inverter with gate driving transformer
- Power transfer up to 150W within the distance of 20cm and achieved maximum efficiency of 90%
- Presented the final report to the department with hardware model

Induction Heater Model Design (Nov 2021 - Dec 2021)

- Prepared the hardware model compatible with market standard coils
- Presented the hardware prototype at ‘Energy Hackathon 4.0’
- Achieved ZVS by resonance oscillation of the LC tank
- Power transfer up to 36W with efficiency of 70% compared to resistor heater

Inductive Wireless Phone Charger (Feb 2019 - Mar 2019)

- PCB embedded pancake coil designed for both transmitter and receiver
- Colpitts oscillator designed with class-A amplifier to feed the coil
- Maximum power transfer achieved up to 5W

GSM based GPS Tracker for Electric Vehicle (Jan 2021 - Mar 2021)

- GPS module and SIM9000 GSM module used with Arduino
- Auto SMS reading and replying feature with encoded request protection

LCR Meter (Oct 2020 - Nov 2021)

- L, C & R measured in different ports for each type of components
- LCD display interfaced with AVR to calculate and display the component values

Spherical Hybrid Black Body Solar (Feb 2018 - Mar 2018)

- Utilized parabolic reflector to concentrate the light rays to array of spherically arranged 12V solar panels
- Solar insolation increment up to 50% was achieved

Participations and Awards

Efficiency Theme Hackathon Winner: Energy Hackathon 4.0 - LOCUS (Dec 2021)

- National level hackathon organized by LOCUS
- Detailed analysis of ‘Transition from LPG to Induction Cooker in Nepal’ was done and presented
- About 30 teams participated within the total of five different themes

Participations: LOCUS Exhibition 2018, 2019 & 2020 - LOCUS

- National level project exhibition and competition organized by LOCUS every year
- Presented the hardware model of ‘Spherical Hybrid Black Body Solar’ at LOCUS 2018

- Presented ‘Inductive Wireless Phone Charger’ at LOCUS 2019
- Designed and Presented ‘LCR Meter’ at LOCUS 2021
- More than 100 teams participated every year

Participation: e-Yantra Robotics Competition (eYRC 2019-20)

- International level robot building competition organized by IIT Bombay
- Developed an automatic object picking and transporting robot within predefined routes

Honorable Mention: NASA Space Apps Challenge - NASA (Oct 2020)

- International level online apps challenge
- Developed ‘Solar Sailer’, space exploration simulation tool with simulated solar system and effects like time dilation, doppler effect and view distortion

Overall and Health Theme Hackathon Winner: Quantum Hackathon - NxtGen (Sep 2020)

- National level online hackathon organized by NxtGen
- Worked on and presented ‘Masked’, a python-based face mask detecting application

Skills

1. Hardware Design: Circuit design, assembly & troubleshooting
2. Equipment/Masurement: Curve tracer, Oscilloscope, Helium based cryocooler
3. Simulation Tools: MATLAB simulink, PLECS, LTSpice, Saber, Proteus, Multisim, Circuit Wizard
4. Circuit Fabrication: Altium Designer, KiCAD, EasyEDA, Circuit Wizard
5. FEA Tools: COMSOL
6. CAD Tools: SOLIDWORKS, AutoCAD
7. Microcontrollers: TI DSP, STM32, AVR, Arduino
8. Programming: MATLAB, MATLAB GUI, C, C++, Java, Embedded C

Publications

- T. Qiu, Z. Zhang, P. Khadka, A. Siraj and D. Rana, "Automated Characterization Platform for Comprehensive Dynamic Rdson Assessment of GaN HEMTs From 50 K to 400 K," 2025 IEEE Applied Power Electronics Conference and Exposition (APEC), Atlanta, GA, USA, 2025 [open↗](#)
- K. Abinash Man, S. Anjana, R. Dilip and R. Ganesh, “INDUCTIVE POWER TRANSFER: ELECTRIC SCOOTER BASED DESIGN”, Final Year Project Report, Apr 2022 [open↗](#)

Research Interests

- Wide Bandgap Power Semiconductors (GaN, SiC)
- Reliability Assessment of Power Semiconductors
- AC/DC Power Converters
- Electric Vehicles and Motor Controllers
- Wireless Power Transmission
- Energy Storage and Transfer System
- Power Electronics and Hardware Design

Involvements

- MATLAB Simulink Workshop (Nov 2021): Lecturer
- Basic Electronics and Proteus Workshop (Jan 2022): Lecturer
- Electrical Club of Pulchowk Campus (2020): Event Manager